Design Document

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1 System overview and institution placement

We propose an exchange that serves all nine settlements through local gateways, with one clearinghouse on Earth and an official price source on Ceres. Clients can transfer NeoDollars, trade Ares Habitat shares, and enter cash-settled futures contracts. Our central rule is simple: reserve the assets before sending an instruction, and keep them reserved until the financial outcome is known.

Distance makes this necessary. Light takes 8.317 minutes to travel one AU. Packets can be lost, paths can pass too close to the Sun, and links can close for maintenance or an incident. A client should not have to manage those conditions. Each settlement therefore has a gateway that accepts local requests in one second, without loss or packet-quota cost in the baseline model. The gateway handles interplanetary communication for the client. The brief fixes the network and funding limits; our institution placement, products, fees, record format, margin terms, queue policy, and packet allocations are design choices.

One authority for ownership. Earth holds the master ledger. Local gateways keep spendable balances and custody records, but they cannot independently finalize a cross-settlement transfer or trade. This avoids repeated rounds of agreement among distant ledgers. Earth can make a financial decision before a remote gateway learns it; the remote client must wait for the final instruction before spending the resulting assets.

Earth has two institutions: the local gateway O_Earth and the clearinghouse C_Earth. Their local handoff takes one second. O_Earth primarily uses Relay A for transfers and share trades; C_Earth primarily uses Relay B for derivative control and settlement. Separate sessions prevent a lost spot packet from blocking later margin messages within that same session. They do not create extra system quota or remove congestion on shared links.

Why Earth and Ceres have different roles. Our draft's 200-Julian-year comparison favors Earth, reporting an 85.77-minute mean one-way delay and a 264.27-minute worst-delay comparison measure. These are design calculations under that comparison, not brief-supplied constants or proof of global optimality. Ceres is attractive at the opening epoch but moves slowly relative to the relays: its approximately 139.62-year relative cycle can leave it on the far side of the relay ring for decades. We use Ceres for the official price observations, rather than making it the permanent ownership authority.

Relay A and Relay B orbit at approximately 2.8284 AU and remain 90 degrees apart. Their separation gives settlements an alternative when one relay leg is blocked by the 0.10-AU solar exclusion region. This geometric coverage argument is separate from complete route availability: the sender still checks the moving receiver, every link, and the flight interval before launch. Packet loss and incidents can delay delivery even when a route is geometrically open.

The model epoch is 2026-09-22 00:00:00 TDB, JD 2461305.5; the fictional date label is 2126-09-22. The service and example tables use elapsed hours from that epoch. Conditional timing means that all required packets arrive; it is not an unconditional deadline.

2 Institutions and opening funding

We charter eleven institutions. The nine gateways provide the same local custody and admission rules. Earth and Ceres each also host a separate specialist institution. A relay assignment identifies the normal session route, not permission to launch through a closure.

Table 1 Chartered institutions

ID	Institution	Settlement	Relay	Responsibility
I-01	O_Mercury	Mercury	A	Local client access and custody of spendable and locked assets.
I-02	O_Venus	Venus	A	Local client access and custody of spendable and locked assets.
I-03	O_Earth	Earth	A	Local custody; Relay A handoff for the clearinghouse.
I-04	O_Mars	Mars	A	Local custody for the seller, factory, and other clients.
I-05	O_Ceres	Ceres	A	Local custody; forwards authenticated observations from the designated price source.
I-06	O_Jupiter	Jupiter	B	Local custody; the faster normal opening-epoch route uses B.
I-07	O_Saturn	Saturn	A	Local client access and custody of spendable and locked assets.
I-08	O_Uranus	Uranus	A	Local client access and custody of spendable and locked assets.
I-09	O_Neptune	Neptune	A	Local custody; avoids the opening B–Neptune maintenance window.
I-10	C_Earth	Earth	B	Master ledger, matching, margin decisions, and final settlement authority.
I-11	C_Ceres	Ceres	B	Official signed price observations for contracts that designate this source; uses O_Ceres to forward through A.

All institutions start with zero NeoDollars and zero shares. The five clients supply $220,000 and 1,200 shares, within the brief's $500,000 and 5,000-share limits. Custody does not make those assets the institution's own capital.

Table 2 Opening client balance sheet

ID	Principal	Settlement	Cash	AH shares	Worked role
A-01	Earth Investor	Earth	$100,000	0	Buys 100 shares from Mars Seller.
A-02	Mars Seller	Mars	$0	1,000	Sells 100 shares at $100 each.
A-03	Mars Factory	Mars	$50,000	0	Takes the long side in the worked metal-futures example.
A-04	Ceres Miner	Ceres	$50,000	0	Takes the short side in the worked metal-futures example.
A-05	Neptune Saver	Neptune	$20,000	200	Transfers $20,000 to an Earth account.
Total	Five clients	Four settlements	$220,000	1,200	Institutions begin with no endowment.

An asset supports only one use at a time. A clearinghouse claim and the collateral backing it are not two available assets. After ownership changes on Earth, a delayed source-gateway record is an unavailable mirror until confirmation retires it. We count its backing once.

Under our fee policy, completed share trades and futures settlements pay a $10 clearing fee from client assets into C_Earth's reserve. Transfers have no clearing fee. Institutional capital can come only from earned fees or explicit contributions debited from the declared opening accounts through our transfer procedure; neither communication failure nor default creates new money.

3 Instruction format and account states

Each network packet is at most 1,024 bytes: a 64-byte transport header and up to 960 bytes of payload. Our uncompressed FIR-64 record occupies 64 bytes, so a packet can carry up to 15 compatible instructions. Each record identifies the operation, its source, and its intended financial effect. Amounts use integer cents.

Table 3 Financial instruction record

Bytes	Field	Width	Meaning
0–1	msg_type	2	0x01 ORDER_LOCK; 0x02 CLEAR_FINAL; 0x03 CANCEL_UNLOCK; 0x04 PRICE_OBS; 0x05 MARGIN_MARK; 0x06 SETTLE_PAY.
2–9	tx_id	8	Unique operation ID: node 8 bits, era 16 bits, sequence 40 bits.
10–17	contract_id	8	Product or contract: 0 for cash, 1 for AH, 2 for the worked futures contract.
18–25	emit_time_ms	8	Elapsed time when the record was issued.
26–33	expiry_time_ms	8	Instruction expiry; -1 denotes a continuing obligation.
34–37	src_principal_node	4	Source principal and settlement, 16 bits each.
38–41	dst_principal_node	4	Destination principal and settlement, 16 bits each.
42–49	quantity	8	Signed share or contract quantity.
50–57	amount_cents	8	NeoDollar amount or price in cents.
58–63	auth_mac48	6	Authentication tag under the brief's unforgeable-principal abstraction.

Transport packet IDs and financial operation IDs serve different purposes. A retry or replacement session can use a new network packet ID while retaining the same tx_id. Earth applies the financial change once and returns the recorded result when an instruction is repeated.

Table 4 Account states

State	Meaning	Who can resolve it
SPENDABLE	Available for withdrawal, transfer, or a new order.	The local gateway checks availability before reserving it.
ENCUMBERED_ESCROW	Reserved for a pending transfer or share trade.	Earth's final or cancellation instruction; a never-dispatched order can be cancelled locally.
ENCUMBERED_MARGIN	Collateral backing an open price-dependent obligation.	Earth's funded settlement or liquidation instruction.

For example, a client who reserves $10,000 cannot use that money for a second order. The gateway rejects the second instruction locally if the remaining spendable balance is insufficient. Reservations and dispatch status are written durably before launch.

An order's expiry limits whether an unexecuted order may proceed. It does not reverse an executed trade or erase an open contract. Once an order has been dispatched, a gateway releases its reservation only after Earth's authenticated outcome. Silence, session expiry, and packet expiry do not authorize release.

4 Worked transfers and share trades

These examples show financial stages, not guaranteed arrival times. Backbone communication includes any needed handshake, launch checks, and confirmations. Assets remain reserved throughout.

Cash transfer from Neptune to Earth

Neptune Saver moves $20,000 to an Earth balance under the same named principal. This destination ledger entry starts empty and does not introduce an extra client identity or opening funds. O_Neptune locks the available money and sends ORDER_LOCK through A and O_Earth to C_Earth. The opening one-way route takes about 250.48 minutes, before handshake or confirmation delays.

Table 5 Transfer lifecycle

Stage	Actor and local knowledge	Action	Financial result
Local request	O_Neptune knows the client has $20,000 available.	Accept locally in one second; reserve funds and send ORDER_LOCK.	Neptune spendable cash becomes $0; $20,000 is in escrow.
Clearing	Earth receives and verifies the reservation and tx_id.	Record the source debit and destination credit together; issue CLEAR_FINAL.	The Earth account has a backed $20,000 claim. The source mirror remains unavailable.
Destination final	O_Earth receives the authenticated final instruction locally.	Apply it once and release the destination balance.	$20,000 becomes spendable on Earth.
Source final	O_Neptune receives the return confirmation.	Retire the old escrow record.	Both sides agree with the authoritative transfer; the operation is complete.

Earth can release the destination balance before Neptune receives confirmation. The source record remains locked. Full completion requires destination spendability and source reconciliation.

Equity trade between Earth and Mars

Earth Investor buys 100 AH shares at $100 each. The buyer reserves $10,010 including the fee; Mars Seller reserves 100 shares. Earth verifies both reservations before exchanging ownership together, providing delivery versus payment.

Table 6 Equity lifecycle

Stage	Actor and local knowledge	Action	Financial result
Buyer reservation	O_Earth verifies the buyer's available cash.	Reserve $10,010 and notify C_Earth locally.	Buyer has $89,990 spendable cash; the purchase and fee are funded.
Seller reservation	O_Mars verifies 100 available shares.	Reserve the shares; send ORDER_LOCK through A.	Seller has 900 available shares and 100 in escrow.
Atomic match	Earth has both valid reservations.	Exchange ownership and assign the $10 fee.	Buyer owns a backed 100-share claim; seller owns a backed $10,000 claim.
Local finals	Each gateway receives its final instruction.	Make the received asset spendable and retire the matching escrow.	Buyer has 100 spendable shares; seller has $10,000 cash and 900 shares; Earth earns $10.

Matching discharges the exchange obligation. Local spendability follows each final instruction; a delayed final does not reverse confirmed ownership.

5 Futures contracts and the worked margin rule

A futures contract binds a long side and a short side to a stated underlying price and maturity. Our design uses cash settlement: the parties exchange the resulting gain or loss instead of delivering the underlying asset. The underlying is a contract term, not a restriction to metals. Each contract must specify an authenticated price source, observation schedule, quantity, multiplier, maturity, fees, and funded margin rules before it opens. Using another underlying requires suitable price data and explicitly agreed terms; it does not change the reservation, ownership, or recovery protocol.

For quantity Q, multiplier m, entry price P_0, and an official observed price P, the cumulative value to the long side is $V = Qm(P - P_0)$; the short side has the opposite value. Both sides post collateral. Our demonstrated futures have an explicitly limited-recourse default rule; this funding restriction must be disclosed in the terms and is not a universal property of futures. A price observation changes their backed claims, while Earth remains the settlement authority. Margin thresholds, call deadlines, and liability caps belong to the particular contract. The numbers below specify our worked example, rather than every futures contract.

Our worked example uses a metal price supplied by Ceres. It pairs Mars Factory, which benefits from higher metal prices, with Ceres Miner, which benefits from lower prices. The contract settles in money rather than requiring metal delivery. We declare the same valuation and margin procedure before either price path is observed.

Table 7 Worked futures terms

Term	Value
Quantity and multiplier	10 contracts × 100 units per contract = 1,000 units.
Entry price and notional	$100 per unit; $100,000 notional.
Initial margin	$25,000 per party, reserved before opening.
Maintenance equity	$10,000 per party.
Example observations	Every 240 hours; the worked run uses signed Ceres prices at hours 0, 240, and 480.
Maturity	Hour 480, twenty days after the scenario starts.
Margin-call response	Restore remaining equity to $25,000 within 24 hours of gateway receipt, using existing available cash.
Liability limit	The losing party's posted collateral caps the funded payment.

The worked contract uses the hour-480 observation for its lifetime payoff if it reaches maturity; hour-240 marking governs margin only. Funded early closure follows the default rule below.

The position opens when Earth has both authenticated margin reservations, the agreed entry-price information, and the complete contract terms, and records the contract. A customer request, an oracle publication, or a packet in flight cannot open it on its own.

For a received price P, the cumulative value owed to the long side is:

V = 1,000 × (P - 100).

The long side's remaining equity is its posted margin plus V; the short side's is its posted margin minus V. A positive V means the miner owes the factory. A negative V means the factory owes the miner. Earth uses received observations only, without knowing later prices.

If the losing side's remaining equity is positive but below $10,000, Earth issues a margin call. The required top-up restores equity to $25,000. The timer begins when the call reaches the losing client's gateway, not when Ceres publishes the price. Additional collateral must be available client money and must be locked and confirmed through the same funding procedure. The gateway records the call-receipt time and any timely top-up durably and sends authenticated evidence to Earth. Earth does not infer receipt or a missed deadline from silence; it waits for that evidence before closing for non-response. Reserved collateral remains locked during this reconciliation.

If remaining equity is zero or negative, Earth closes the contract and settles at the posted-collateral cap. This default rule takes priority over a margin call at zero equity. After an unmet call is established, Earth closes using the latest applicable received price. Payment is the current loss capped by the losing

side's posted collateral, not automatic forfeiture of the entire margin. There is no unlimited residual debt under this contract.

Interim marking assigns backed claims within the reserved margin. It does not release the winner's interim gain for spending. For a contract reaching maturity without default, Earth pays the final total once: the hour-240 mark plus the final increment equals 1,000 × (P_480 - 100). The winner pays the $10 settlement fee from released cash. If the final gain is zero, the contract names the fee payer before opening.

The worked contract has a fixed hour-480 maturity. A later contract must name its own maturity and official pricing inputs, receive a new identifier, and obtain fresh funded reservations. Ending a contract or a test window does not automatically renew it or reset any balances.

6 Worked price paths and the binding constraint

The rising and falling paths are independent runs, each explicitly reset to the full opening balance sheet. Both start from the same balances and $25,000 margin on each side. The following paths are test scripts, not prices known in advance. Our actors behave identically until they receive different observations.

Table 8 Symmetric response to rising and falling prices

Stage	Rising path	Falling path
Opening price	$100; V = $0. Both parties reserve $25,000.	$100; V = $0. Both parties reserve $25,000.
Hour-240 observation	Price $120; V = +$20,000. Miner equity is $5,000.	Price $80; V = -$20,000. Factory equity is $5,000.
Received margin call	Miner reserves another $20,000, taking posted margin to $45,000 and remaining equity to $25,000.	Factory reserves another $20,000, taking posted margin to $45,000 and remaining equity to $25,000.
Hour-480 observation	Price $125; total gain to factory is $25,000.	Price $75; total gain to miner is $25,000.
Final remaining equity	Miner has $45,000 - $25,000 = $20,000; no funded default after the successful top-up.	Factory has $45,000 - $25,000 = $20,000; no funded default after the successful top-up.
Fully completed balances	Factory $74,990; miner $25,000; clearinghouse $10.	Factory $25,000; miner $74,990; clearinghouse $10.

For example, in the rising run the $20,000 interim claim and the final $5,000 increase produce one $25,000 gain. Earth must not pay $20,000 and then another $25,000. The winner receives its own $25,000 margin back plus the $25,000 gain, less the fee. The losing party receives its unused $20,000 margin; its other $5,000 remained available after the top-up.

Default depends on actual posted collateral, not on a fixed percentage price move after a top-up. A move to $125 exhausts an unchanged $25,000 reserve, but does not exhaust a reserve that has grown to $45,000. The same distinction applies to the falling path.

Variation that makes funding bind. In a separate run reset to the full opening balance sheet, declare $15,000 initial margin on each side before opening and retain the same 1,000-unit exposure. A rise from $100 to $120 creates a nominal $20,000 loss for the miner. Its remaining equity is -$5,000, so Earth closes the contract. The factory receives the funded $15,000 cap; the additional $5,000 is not paid and does not become an unbacked claim. A fall to $80 applies the same rule with the roles exchanged.

In the standard runs, the successful top-up raises combined encumbered margin from $50,000 to $70,000. That is 31.82% of the $220,000 opening cash pool, above the required 25% utilization threshold. No new cash is supplied to reach that scale. Cash and share encumbrances are measured separately. Asset-hours follow the reserved backing across authoritative reassignment until release, without adding claims or source mirrors as extra assets.

Maturity is not immediate access to cash. The hour-480 publication first travels from C_Ceres to O_Ceres locally, then through A to Earth. Earth makes the settlement decision on receipt. Its final

instructions travel to the participants, whose gateways release the funded balances. The contract's maturity, Earth's financial decision, and each client's spendability are separate moments.

7 Product guarantees and service conditions

All nine settlements can access the three worked products through their local gateway. A client's location changes the communication delay and route, not the requirement to fund a trade. The product table separates the demonstrated procedures from the additional framework and future extensions.

Table 9 Product scope and funding conditions

Product	Status	Funding and end of life
NeoDollar transfers	Implemented	Transfer existing cash one for one. Reserve at the source, reassign on Earth, and release at the destination on the final instruction. No clearing fee.
Ares Habitat equity	Implemented	Reserve purchase cash and fee plus the seller's shares. Earth exchanges ownership atomically. Both local finals must be applied for full completion.
Cash-settled futures	Implemented	Each contract names its underlying price, official source, quantity, multiplier, maturity, observation schedule, and margin terms. Reserve margin on both sides and use authenticated observations received by Earth. The worked metal contract uses Ceres prices. Default is capped by the loser's posted collateral.
European capped options and short positions	Framework	Terms identify the payoff and price source. Writers reserve the maximum capped payment; covered or capped shorts use our proposed 150% entry-value collateral rule. That percentage alone does not fund unlimited losses: admission also requires sufficient reserved backing for the contractual maximum loss, or custody of the shares for a covered short. Earth settles or liquidates at the cap.
Zero-coupon bonds and collateralized lending	Extension	A loan creates matching claims and liabilities. Our proposed collateral threshold is 125% of principal plus the fixed repayment fee; this is an extension term, not a brief requirement. Funded default transfers the pledged collateral and ends the obligation under a declared limited-recourse rule. This is not a promise that its market value cannot fall.
Synthetic settlement currencies	Extension	A local token is backed one for one by existing reserved NeoDollars. Issuance creates a claim and liability; it does not create additional backing.

Financial guarantees. Verified assets are reserved before an order leaves its gateway. The master ledger changes ownership once per operation. Share trades exchange both sides together. Derivative payments stay within funded collateral. A delayed instruction cannot make reserved assets available for a second use.

These guarantees rely on the brief's unforgeable identities, durable financial records, and rule-following operators, together with our gateway reservation procedure. Malicious or colluding operators are outside the demonstrated baseline. The FIR-64 authentication field represents this identity abstraction; it is not a claim that a 48-bit tag establishes real-world cryptographic security. We do not promise unlimited derivative payouts, credit during an outage, or spendability at a remote settlement before its gateway receives the necessary instruction.

Service conditions. A gateway admits an order only when assets, local queue space, and a valid operation identifier are available. It dispatches only when its packet allowance, session, route, and launch checks permit. During a closure it may retain an admitted order, reject additional spot activity, or wait to resolve a cancellation. The customer's reserved assets remain unavailable during that wait.

The transport can lose an arbitrary sequence of packets. Consequently, no finite completion deadline or recovery time is guaranteed unconditionally. Progress requires eventual delivery of the necessary price and financial messages, valid sessions, and available capacity; until then, unresolved obligations keep their named funded owner. The evidence appendix reports conditional traces, launch probabilities, traffic, and capital measures. A faster one-way path is useful, but does not by itself establish faster financial completion.

Transfers and equities require Earth's ownership authority. Futures require authenticated observations from the price source named in their contract, delivered to Earth before valuation. Ceres supplies those observations in our worked example; this does not restrict futures to metals. The options framework names its valuation source in the contract. We do not treat an observation as freely broadcast to every settlement.

8 Service at every settlement

Table 10 retains the design's normal opening routes and hour-300 comparison. Delays are one-way minutes to Earth's clearing authority; Earth's local handoff is one second. Availability is the fraction of the next 24 hours when every link on the listed route can launch, not the probability that a packet arrives. The figures describe those sampled windows and do not promise availability at every later epoch.

Table 10 Normal service routes and restrictions

Settlement	Relay	Delay 0 / 300	Launch availability 0 / 300	Products	Local restriction or fallback
Mercury	A	46.02 / 43.77	100% / 100%	T E F; O S	Use a B session when the A leg is solar-blocked.
Venus	A	40.57 / 37.67	100% / 100%	T E F; O S	Use a B session when the A leg is solar-blocked.
Earth	Local	0.02 / 0.02	100% / 100%	T E F; O S	Local handoff persists; isolation prevents remote clearing progress.
Mars	A	33.91 / 33.44	100% / 100%	T E F; O S	Use B when the A leg is unavailable.
Ceres	A	33.37 / 32.24	100% / 100%	T E F; O S	Forward prices through A during B–Ceres maintenance. Return calls through B can wait.
Jupiter	B	50.92 / 50.23	100% / 100%	T E F; O S	B is 16.2 minutes faster than A at the opening epoch.
Saturn	A	79.42 / 78.86	100% / 100%	T E F; O S	Long flights delay confirmations; a margin-call timer starts at local receipt.
Uranus	A	158.25 / 156.72	100% / 100%	T E F; O S	High launch loss; retries and unresolved-escrow recovery apply.
Neptune	A	250.48 / 250.05	100% / 100%	T E F; O S	Avoid B–Neptune maintenance at hours 2–26; retain escrow if retries fail.

T = cash transfer; E = equity; F = futures. O = capped options and S = short positions under the additional framework. The funding conditions in Table 9 apply at every settlement. Local client access is one second and lossless throughout the common baseline.

Direction matters. Spot orders and Ceres observations normally travel through A and O_Earth, then enter C_Earth locally. C_Earth's derivative calls and settlement instructions normally travel through B. Jupiter uses B for its inbound clearing route. The table describes incoming access to clearing; reverse messages and price delivery are additional dependencies in a complete transaction.

In the worked metal-futures scenario, at hour 240, C_Ceres releases the observation locally and O_Ceres forwards it through A. This avoids B–Ceres maintenance for the price report itself. If the miner owes

a margin top-up, however, Earth's outgoing call uses B and may wait until hour 264. The 24-hour response period starts when O_Ceres actually receives that call. A Mars-bound call is not blocked by maintenance on the Ceres leg.

Every route is rechecked at dispatch. Within a session the route remains pinned; changing relays requires a suitable surviving or replacement session. A known maintenance window prevents launches whose flights would overlap it, including launches shortly before the window starts. A geometrically available alternative still consumes link capacity and can suffer random loss.

Initial route rankings can change as bodies move. The longer-horizon evidence therefore assesses orbital phase as well as the opening geometry. Those comparisons do not reset ongoing financial state or replay the one-time maintenance windows.

9 Transport failures and financial recovery

The backbone first establishes a session using SYN, SYN-ACK, and the final ACK. Each directed link serializes one packet per second and has a 10,000-packet FIFO queue. Each peer connection permits at most 64 unacknowledged DATA packets. Relays add one second of processing before a received packet is ready for its next launch; simultaneous arrivals use the brief's sender-node and packet-sequence ordering. A receiver releases DATA only after the handshake completes. A hop receipt and an endpoint DATA acknowledgment have different jobs.

The hop timer is R_h = twice the predicted hop flight time plus one hour; at most four launches are allowed. The endpoint timer is $R_e = 2T_0 + 24$ hours. At each enqueue, T_0 includes moving-receiver flight, serialization, and relay processing with empty queues and no closure waiting; a blocked link uses its pure light time as if open. SYN, SYN-ACK, and each DATA packet have at most four endpoint attempts; the final ACK has no timer of its own. These timers overlap. An endpoint attempt counts when enqueued, even if it cannot launch; a hop attempt counts only at launch. A due endpoint retry replaces any earlier unlaunched copy. A retry receives a new packet ID but retains its session byte sequence and financial operation ID.

Table 11 Local knowledge and obligations under failure

Event	What the sender or gateway knows and does	Earth and the financial obligation
Forward packet lost	No receipt proves neither success nor failure. Retry under the transport timers.	Earth acts only on received data. Escrow or margin remains locked.
Receipt lost after arrival	Sender may retry. Receiver recognizes the duplicate packet ID and returns a fresh receipt without forwarding again.	An existing decision remains valid. A lost receipt cannot reverse it.
Hop or endpoint attempts exhausted	Delivery status is unknown. Retain the reservation and recover through an application resubmission with the same tx_id.	Earth returns an existing result, executes a valid unprocessed instruction, or authorizes cancellation of an expired unexecuted order.
Duplicate financial instruction	Transport and application records identify the repeated operation.	Earth applies no second debit, credit, or payment; it returns the recorded result.
Contradictory client order	Gateway checks spendable assets. A second use of already reserved assets is rejected locally.	No second funded order reaches clearing.
Endpoint reset or idle expiry	Dead-session packets are ignored. Establish a fresh session or use a valid surviving endpoint.	Durable balances, claims, and obligations survive. Resume without repeating their financial effect.
Gateway isolation	Keep unresolved work and reservations durably. Dispatch or reconcile after reconnection.	Already recorded ownership remains authoritative; pending finals still need delivery before local spending.

The seven-day idle rule matters. A session expires after 168 hours without receiving a session packet. Our 240-hour observation gaps are longer, so a quiet connection can expire between marks. Unless other received traffic kept it alive, the next report needs a new handshake. Its SYN and other quota-counted traffic must fit the packet budget. Maintaining a financial contract does not keep a transport session alive.

The original network packet and all its automatic hop and endpoint retries share one 30-day lifetime from original creation. A new application resubmission uses a fresh network packet and lifetime, keeps the same financial operation ID, and counts as new quota traffic. That expiry stops transport, not the underlying obligation. After reconnection, recovery is complete only when the needed decisions have been applied, the recipients can spend their funded balances, and obsolete source records are retired. Receiving one recovery packet is not enough.

10 Ranked risks and the stress case

Our main risks arise where the financial procedure meets the physical and transport rules. We rank them below and state what the treatment does and where it stops.

Table 12 Architectural risks and treatments

Rank	Risk and why it matters	Treatment and remaining limit
1 Critical	Ceres and the relays have a roughly 139.62-year relative cycle. Its initially favorable placement can deteriorate for decades.	Keep Earth as clearing authority and Ceres as price source. Compare later orbital phases rather than choosing the hub from hour 0 alone.
2 High	B–Ceres maintenance begins at the hour-240 mark. A usable incoming price route does not guarantee an immediate outgoing margin call.	Forward the observation through A. Keep funded margin reserved while B-bound instructions wait; start the response clock at receipt.
3 High	In-order session delivery can hold urgent messages behind a lost routine message.	Separate spot sessions on O_Earth/A from derivative sessions on C_Earth/B. Shared link queues and the system quota remain common constraints.
4 Medium	Neptune's long relay leg has high loss and can exhaust automatic attempts.	Preserve reservations, retry, and use application recovery with the same operation ID. There is no unconditional completion deadline.
5 Medium	Endpoint reset drops sessions and can invalidate in-flight traffic addressed to them.	Use surviving institutional endpoints where possible and rebuild the affected sessions. Ledger records and operation IDs survive; financial recovery is not a one-second handoff.

For Neptune's opening A leg, the design uses a distance near 27.86 AU and per-launch loss p ≈ 42.72%, from p = 1 - exp(-0.02d). Under independent forward losses at fixed distance, all four forward launches fail with probability p^4 ≈ 3.33%. Sixteen such failed launches give an idealized probability near 1.23 × 10^-6.

That calculation explains the benefit of retries, but is not a 99.99988% guarantee of completed financial service. Receipts, handshakes, multiple route legs, queues, session state, and closure timing also matter. The full protocol must be assessed at actual launch times. The direct service has a different loss formula, 1 - exp(-0.08d), and no automatic retry protection.

Why isolate Earth for the main recovery test. A 72-hour Earth gateway isolation while futures settlement messages are pending targets our single clearing dependency. Both relay choices lose access to the same authority. A forced-loss incident attacks particular launches, while an endpoint reset can leave the second Earth institution's sessions available. Isolation therefore tests whether we preserve obligations when neither alternate routing nor a surviving endpoint can immediately restore remote service.

During isolation, an undecided instruction remains reserved; an already settled claim remains au-

thoritative but may not yet be spendable remotely. After reconnection, we reconcile the same operations and complete their final instructions. The comparison with maintenance removed separates scheduled closure from the incident. A secondary forced-loss sensitivity at the hour-240 mark tests retry behavior while B–Ceres is unavailable.

The fourth endpoint attempt occurs three R_e intervals after the original attempt, not necessarily three intervals after an incident begins. It can fall after a 72-hour outage in the aligned timing example, but it is not a universal recovery theorem. Route availability, random loss, and valid sessions still determine whether it succeeds.

11 Packet allowances and excess demand

The backbone allows 600 newly originated packets across the system in every rolling 24-hour window. Distant gateways cannot consult an instantaneous global counter, so the updated design assigns allowances in advance. Each principal enforces its local allowance from durable origination timestamps, including after a restart.

Table 13 Backbone allocation

Holder or purpose	Packets per rolling 24 hours
Eight non-Earth gateways	35 each; 280 combined.
C_Earth outbound instructions	280.
C_Ceres price-source allocation	15.
Shared emergency reserve at O_Earth and C_Earth	25 combined, not 25 for each institution.
System total	600.

The 280/280 pairing reserves room for replies alongside incoming work. It does not guarantee that every financial reply can launch immediately: requests from different origination windows can arrive together, and a transaction can require more than one control message. Reply work queues when necessary. New DATA, application resubmissions, application control, and SYN traffic count. Automatic retries and the brief's exempt transport acknowledgments still use link capacity and count in communication efficiency.

A price forwarded by O_Ceres is charged to O_Ceres's allowance. The separate oracle allocation cannot be used to relabel another principal's transmission. Official financial coordination uses the backbone. Direct service has its own twelve-packet rolling allowance per principal, sixty-second launch separation, and no automatic retries. Two Ceres institutions have separate allowances, not a pooled allowance for one sender.

When demand exceeds available packet capacity, we use the following procedure.

1. **Batch waiting work.** After a gateway uses more than twenty local allowances in a window, pack up to fifteen compatible FIR-64 records per packet. Earth can batch confirmations addressed to the same gateway.

2. **Request extra allowance.** Piggyback REQUEST_EXTRA_QUOTA while outbound capacity remains. Earth grants additional inbound and reply reservations in matched pairs. A gateway does not assume that a quiet peer's allowance is free. Reassignment must be confirmed, must retire the previous permission, and must respect the previous holder's rolling launch history. If safe transfer cannot be established, the request waits.

3. **Keep a local priority queue.** Price marks, margin top-ups, and liquidations precede new spot orders and transfers. Equal-priority requests retain FIFO order. Collateral stays reserved while an admitted instruction waits.

4. **Bound admission.** At 1,000 queued instructions, reject additional spot requests locally in one second and release any reservation for the rejected, never-dispatched request.

An order that has never been sent to Earth can expire or be cancelled locally in one second without a backbone packet. Its durable dispatch record must prove that no copy or local handoff reached clearing. Once dispatched, cancellation needs Earth's authenticated decision.

Batching fifteen records reduces packet use; it does not imply fifteen complete transactions per packet or a fixed daily transaction rate. Throughput depends on compatible demand and the actual control, session, and settlement traffic. Accepted work remains bounded by the packet allocations and queue limits; overload waits or is rejected.

12 Continuing operation and deployment limits

The exchange has no planned end time. Scenario windows, packet lifetimes, and orbital scans do not reset it. Open contracts retain funded owners until resolved. Later-maturing obligations use new transport packets with the same financial identifier. Ceres continues the 240-hour publication schedule used by the worked example. Each later futures contract names its underlying price, designated source, maturity, and applicable observations before opening; the example schedule and hour-480 maturity do not define futures generally.

After a long outage, gateways reconcile with Earth before releasing reservations. Valid sessions, allowances, and batching drain the backlog. A 64-packet window holds up to 960 instruction records; two Earth sessions can provide 128 DATA slots. Complete transactions still require their full instruction sequences and share the system quota.

Network duplicate records cover the packet lifetime and thirty days after its last receipt, as the brief requires. Under our long-term storage policy, completed operations older than a year can be compacted into per-account balance checkpoints, while preserving sequence information needed to reject old replays and all unresolved exceptions. Unresolved obligations are never discarded. Transaction IDs include a forty-bit sequence and sixteen-bit era; advance the era before sequence exhaustion and never reuse an identity. Our session-ID convention uses a persistent node identity and 64-bit monotonic counter; it is a design choice implementing the brief's prohibition on session-identity reuse. Unsafe namespace or storage exhaustion stops admission, not existing obligations.

Earned fees and explicit funded contributions replenish institutional capital. No balance, queue, reserve, or quota resets at midnight or at a test boundary.

Table 14 Physical effects outside the common baseline

Effect	Estimate and stated assumptions	Consequence for the design
Doppler shift and timekeeping	Assume relative speed up to 96.3 km/s: $\Delta f/f \approx 3.21 \times 10^{-4}$; 64.2 GHz at 200 THz or 10.3 MHz at 32 GHz. Illustrative kinematic clock scale at that assumed speed: about 4.46 ms/day. The speed and carrier frequencies are engineering inputs, not brief-supplied maxima.	Requires frequency tracking and consistent timekeeping. These deployment estimates do not add baseline losses. Reservations survive equipment outages.
Solar gravitational delay	Use 0.116 ms only as an illustrative Shapiro-delay scale. It is not an established worst-case bound; endpoints and impact parameter determine the actual correction.	Small beside serialization and retry timers; relevant to the one-millisecond light-time tolerance. No early asset release.
Surface visibility and local relay handoff	For an illustrative deployment assumption, half of a roughly 24.6-hour Mars rotation would give 12.3 hours of obscuration; actual visibility requires station geometry. Assumed local orbital handoff adds 0.3–2.5 seconds.	Local access needs suitable infrastructure. Handoff figures are engineering assumptions outside the brief; custody rules continue during outages.

These effects are outside the common Keplerian baseline. Higher-fidelity equipment and local orbital networks are deployment work; they do not replace the baseline evidence or create a stronger unconditional service promise.